

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M.Pharm (Quality Assurance)

University Theory Examination January 2018

PHQA003PHARMACEUTICAL ANALYSIS-I

Date: 05.01.2018, Friday

Time: 10.00 a.m to 01.00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1 A) Define Calibration and discuss its importance. How is validation different from calibration? Describe calibration of UV-Vis Spectrophotometer. 05
- B) Describe the terms in brief: 1. pKa 2. Partition coefficient 3. Solubility 4. Reference Standard 5. Ash Value. 05
- Q.2 A) Justify: FSSAI is revolutionary in terms of Food Quality and Safety in India 05
- B) What are the tests, would you like to suggest to evaluate quality of fixed oil. 05
- Write significance for each test.

OR

- Q.2 Write a detailed note on Analytical Method Development with special reference to steps involved in developing an analytical method, with suitable case study. 10
- Q.3 Answer the following (Any three) 15
- (A) Write a note on LIMS and its applications.
- (B) Justify the need of mandatory analysis of pesticides in food products. Draw an outline of procedure to be adopted to estimate preservatives.
- (C) Write a note on - Multidisciplinary Aspect of Pharmaceutical Analysis.
- (D) List the tests to be performed to evaluate tablets as per IP. Write significance for each test.

SECTION-II

- Q.4 (A) Write a detailed note on HPLC system calibration (Cite any three parameters) 10
(B) Write a summary of methods proposed in pharmacopoeia to evaluate quality of antidiabetic drugs (write for five such drugs)

- Q.5 A herbal material is given to you, in powder form. Suggest tests to be performed to evaluate quality of the herbal material 10

OR

- Q.5 What is Analytical Method Validation? Write its objectives. Describe following validation parameters with respect to explanation, assessment methodology and acceptance criteria. 10

- Q.6 (1) Specificity (2) Linearity (3) Precision.
Answer the following (Any three) 15

- (A) Write a note on toxicity testing in cosmetics
(B) Make a note on : Evaluation of Creams as topical care product
(C) An SVP showed non compliance in results of chloride content as well as in impurity profile. What might be the sources of error. Suggest other tests to be performed for SVP as per IP.
(D) Classify solid-state properties of drug substances. Exemplify analytical techniques used for solid-state analysis (any two).